Software Development Plan: PrintHive

Project Overview

PrintHive is a web-based system designed to optimize 3D printing at Purdue maker spaces, specifically The Hive, making operations simpler for staff and students. Students can submit raw .stl files which are algorithmically organised by estimated print time, active staffing hours, and printer availability. The system features dynamic batching of compatible prints, focusing on reducing manual intervention and improving how the maker space manages its print queue and hardware fleet.

Scope

The following system boundaries define the core capabilities required to deliver a minimum viable product for the PrintHive platform.

In Scope

- **SYS-01: User & Staff Authentication:** General users will authenticate via email domain verification by entering a code sent to their email. Staff members will authenticate using a secondary SSO system via Purdue Hackers ID (https://github.com/purduehackers/id).
- **SYS-02: Model Validation:** The application will process raw .stl or .obj files and verify that the geometry fits within the selected printer's build volume before queue entry.
- **SYS-03: Model Preview:** Users will be provided a WebGL viewport to inspect their submitted model alongside general slicer settings.
- **SYS-04: Automated Slicing:** A headless Orca Slicer command-line engine will asynchronously process and compile printer-ready G-code.
- **SYS-05: Staff-Shift-Aware Scheduling:** The scheduling engine will correlate estimated job durations with active student-attendant staffing shifts to prevent prints from finishing in an empty lab.
- **SYS-06: Dynamic Batching & Nesting:** The system will algorithmically group compatible models onto a single build plate to maximise hardware utilisation and reduce idle thermal cycles.
- **SYS-07: Real-Time Telemetry Video Feed:** The system will integrate live webcam feeds and printer telemetry into the user dashboard to allow remote monitoring of print progress, printer status, and potential failures.

Under Evaluation

- **SYS-08: Filament Inventory Tracking:** Tracking exact filament spool weights via the database. *Evaluation method:* We will assess if lab attendants have the capacity to manually input spool weights upon changeover to keep the database accurate.
- **SYS-09: Automated AI Failure Detection:** Utilising visual feeds to automatically halt failed "spaghetti" prints. *Evaluation method:* Feasibility depends on the availability of sufficient compute power on the local network edge daemons.

- **SYS-10: Estimated Queue Wait Times:** The system may provide students with estimated wait times for their submitted print jobs based on queue length, estimated print durations, printer availability, and staffing schedules. Evaluation method: We will compare predicted wait times against simulated print schedules to determine whether the estimates are sufficiently accurate and reliable for students.

Out of Scope

- **SYS-11: Custom Slicing Engine Development:** The project will strictly rely on the open-source Orca Slicer CLI rather than developing a proprietary geometric slicing algorithm from scratch.
- **SYS-12: Native Desktop/Mobile Application:** The solution will be delivered exclusively as a responsive web platform to eliminate local installation friction for users.
- **SYS-13: Direct Hardware Modification:** The team will interface with existing Prusa and Bambu Lab APIs rather than flashing custom firmware onto the 3D printers.

Assumptions

- Maker spaces maintain and publish predictable, stable staffing schedules for the scheduling engine to reference.
- Stable local network connectivity exists between the edge relay daemons and the physical 3D printers.

Constraints

- University firewall rules prohibit inbound network traffic, requiring the use of outbound-only encrypted tunnels (e.g., WebSockets or Tailscale).
- System functionality is constrained by the distinct proprietary APIs of different manufacturers (Bambu Lab MQTT vs PrusaLink HTTP).

Requirements Elicitation & Prioritisation

Through collaborative whiteboard sessions, the team applied the MoSCoW method to filter raw brainstorming ideas into actionable system constraints. Categorising features into "Must Have" versus "Could Have" clarified the critical path, ensuring engineering resources are appropriately allocated to the scheduling algorithms and live monitoring feeds.

Software Requirements

The following requirements utilise the Easy Approach to Requirements Syntax (EARS) to provide verifiable specifications.

Functional Requirements

1. **FR-01: User & Staff Authentication** | *Statement:* The system shall authenticate general users via email domain verification using a sent code, and staff members via a secondary SSO system using Purdue Hackers ID. | *Rationale:* Prevents unauthorised network access and secures staff administrative capabilities. | *Test:* Attempt login with invalid email domain or unauthorized staff SSO; verify rejection. | *Context:* Email verification code and Purdue Hackers ID SSO integration. | *Trace:* SYS-01 | *Priority:* Must Have
2. **FR-02: File Upload** | *Statement:* The system shall accept 3D model uploads in .stl and .obj formats up to 50MB. | *Rationale:* Standardises ingestion formats. | *Test:* Upload a 45MB .stl and a 55MB .stl; verify only the former succeeds. | *Context:* Web portal intake. | *Trace:* SYS-02 | *Priority:* Must Have
3. **FR-03: Geometry Validation** | *Statement:* The system shall reject models whose geometric bounding box exceeds the target printer's maximum build volume. | *Rationale:* Prevents physical hardware collisions. | *Test:* Submit a 300x300mm object to a 250x250mm profile; verify error. | *Context:* 3D space calculation. | *Trace:* SYS-02 | *Priority:* Must Have
4. **FR-04: Model Preview** | *Statement:* The system shall render the uploaded 3D mesh within an interactive WebGL viewport in the browser. | *Rationale:* Allows users to visually confirm orientation. | *Test:* Upload valid file; verify browser renders manipulatable 3D object. | *Context:* Client-side rendering. | *Trace:* SYS-03 | *Priority:* Must Have
5. **FR-05: Slicer Profile Selection** | *Statement:* The system shall provide exactly three standard functional profiles (Draft, Standard, High Strength) for user selection. | *Rationale:* Abstracts complex slicer settings from the user. | *Test:* Verify UI restricts input to the three designated profiles. | *Context:* Pre-configured Orca Slicer parameter sets. | *Trace:* SYS-03 | *Priority:* Must Have
6. **FR-06: Slicing Dispatch** | *Statement:* The system shall dispatch validated models to a containerised Orca Slicer CLI worker queue. | *Rationale:* Enables asynchronous background processing. | *Test:* Submit job; verify worker daemon acknowledges receipt within 2 seconds. | *Context:* Redis/Celery queue. | *Trace:* SYS-04 | *Priority:* Must Have
7. **FR-07: G-code Generation** | *Statement:* The slicing worker shall generate printer-specific G-code tailored to the assigned hardware profile. | *Rationale:* Translates geometry into physical machine instructions. | *Test:* Parse output file; verify standard G-code formatting and headers. | *Context:* Automated toolpath compilation. | *Trace:* SYS-04 | *Priority:* Must Have
8. **FR-08: Job Duration Extraction** | *Statement:* The system shall parse the compiled G-code to extract the estimated print duration in minutes. | *Rationale:* Necessary for shift-aware scheduling calculations. | *Test:* Slice standard 60-minute model; verify database registers time within a 5% margin. | *Context:* Metadata reading. | *Trace:* SYS-05 | *Priority:* Must Have
9. **FR-09: Shift Schedule Ingestion** | *Statement:* The system shall ingest and store weekly active staffing shifts for the maker space. | *Rationale:* Establishes the temporal constraints for job dispatch. | *Test:* Input a 08:00–17:00 shift block; verify accurate database entry. | *Context:* Attendant schedules. | *Trace:* SYS-05 | *Priority:* Must Have
10. **FR-10: Shift-Aware Dispatch** | *Statement:* The system shall only dispatch a print job if the extracted job duration is less than the time remaining in the current active staff shift. |

Rationale: Prevents prints from sitting idle on unstaffed heated beds. | *Test:* Submit a 4-hour print with 2 hours left in a shift; verify job remains queued. | *Context:* Core scheduling algorithm. | *Trace:* SYS-05 | *Priority:* Must Have

11. **FR-11: Dynamic Print Batching** | *Statement:* The system shall algorithmically group multiple compatible user models onto a single virtual build plate prior to slicing. | *Rationale:* Maximises throughput and reduces thermal cycle waste. | *Test:* Submit three 50x50mm objects; verify system compiles them into one G-code file. | *Context:* 2D bin-packing logic. | *Trace:* SYS-06 | *Priority:* Must Have
12. **FR-12: Edge Relay Tunnel** | *Statement:* The edge relay daemon shall establish an encrypted, outbound-only WebSocket connection to the central orchestrator. | *Rationale:* Bypasses inbound firewall restrictions securely. | *Test:* Deploy edge relay on restricted subnet; verify bidirectional communication succeeds. | *Context:* Local network bridging. | *Trace:* SYS-07 | *Priority:* Must Have
13. **FR-13: Telemetry Polling** | *Statement:* The edge relay shall poll target printers every 10 seconds for active nozzle and bed temperatures. | *Rationale:* Maintains live fleet status for the central dashboard. | *Test:* Monitor network traffic; verify HTTP GET/MQTT payload retrieval every 10s. | *Context:* Hardware state monitoring. | *Trace:* SYS-07 | *Priority:* Must Have
14. **FR-14: Manual Bed Clearance** | *Statement:* The system shall require a manual "Bed Cleared" boolean trigger from a staff account before dispatching the next queued job to a completed printer. | *Rationale:* Prevents the nozzle from crashing into finished parts. | *Test:* Simulate finished print; verify queue halts until clearance button is clicked. | *Context:* Operational safety interlock. | *Trace:* SYS-05 | *Priority:* Must Have
15. **FR-15: Auto-Requeue on Error** | *Statement:* The system shall automatically return a job to the top of the processing queue if the printer reports a hardware failure code. | *Rationale:* Ensures users do not lose their spot due to mechanical faults. | *Test:* Trigger thermal runaway error on printer; verify job status reverts to 'Queued'. | *Context:* Fault tolerance. | *Trace:* SYS-05 | *Priority:* Must Have
16. **FR-16: Live Video Streaming** | *Statement:* The system shall display a real-time video feed from the target printer on the user dashboard at a minimum of 15 frames per second. | *Rationale:* Enables remote visual verification of print integrity and early failure detection. | *Test:* Connect a compatible webcam and verify the dashboard renders the continuous stream with sub-2-second latency. | *Context:* WebRTC or MJPEG stream integration via the edge relay. | *Trace:* SYS-07 | *Priority:* Must Have
17. **FR-17: Dynamic Priority Scheduling** | *Statement:* When multiple print jobs are eligible for dispatch, the system shall select the job with the highest effective priority based on its initial priority weight and time spent waiting in the queue. Jobs with equal priority scores shall be ordered by submission time and job ID. | *Rationale:* Reduces the possibility of lower priority jobs experiencing indefinite delays while maintaining predictable queue behavior. | *Test:* Submit multiple jobs with different priority weights and arrival times. Verify that the scheduler applies the defined priority formula and consistently selects jobs according to the calculated order. | *Context:* Dynamic priority weighting, linear aging, and deterministic queue ordering. | *Trace:* SYS-05 | *Priority:* Must Have

Non-Functional Requirements

1. **NFR-01: Performance:** *Statement:* The system shall complete headless slicing of a standard 20MB .stl file within 60 seconds under defined reference hardware conditions. | *Rationale:* Ensures students receive processed print jobs without excessive delays. | *Test:* Execute the slicing process on the team's reference hardware using a 20MB test model across 10 trials. Record the processing time of each trial and verify that each completes within 60 seconds. | *Context:* Orca Slicer CLI processing and backend performance. | *Priority:* Must Have
2. **NFR-02: Reliability:** *Statement:* The local edge relay shall preserve active print dispatch state during a temporary external network interruption and restore communication when connectivity returns without issuing duplicate print commands. | *Rationale:* Prevents interrupted network connections from causing duplicate submissions or loss of printer state. | *Test:* Simulate a network disconnection during an active print. Restore connectivity and verify that the relay reconnects, preserves the existing job state, and does not dispatch the same job twice. | *Context:* Non-volatile state storage and edge relay recovery. | *Priority:* Must Have
3. **NFR-03: Security:** *Statement:* The system shall not store user passwords, authentication tokens, or other sensitive credentials in plaintext, and shall restrict administrative operations to authorised staff accounts.| *Rationale:* Protects student information and prevents unauthorised access to printer management functions. | *Test:* Inspect database records, stored configuration files, and authentication handling for plaintext credentials. Attempt administrative operations using a standard student account and verify that access is denied. | *Context:* Authentication security and role-based permissions. | *Priority:* Must Have
4. **NFR-04: Maintainability:** *Statement:* The Python application backend shall use the uv package manager and a version-controlled lockfile to maintain reproducible dependency installations. | *Rationale:* Reduces dependency conflicts and simplifies development across different team environments. | *Test:* Clone the repository into two clean development environments and execute uv sync. Verify that both environments install the dependency versions specified by the committed lockfile. | *Context:* Python dependency management and reproducible development environments. | *Priority:* Must Have
5. **NFR-05: Usability:** *Statement:* The web dashboard shall become interactive within 1.5 seconds under a defined simulated 4G network connection. | *Rationale:* Allows students to access print submission and monitoring features without excessive loading delays. | *Test:* Use browser performance testing tools with a consistent simulated 4G connection to measure dashboard loading time across 10 trials. Verify that the time to interactivity remains within the specified limit. | *Context:* Web application responsiveness and mobile accessibility. | *Priority:* Must Have
6. **NFR-06: Scheduling Evaluation:** *Statement:* The scheduling evaluation framework shall report average job wait time, printer utilisation, and completed job count for both the PrintHive heuristic scheduler and a First-Come, First-Served baseline using identical simulated workloads. | *Rationale:* Provides measurable evidence for determining whether the proposed scheduling approach improves print queue management. | *Test:* Execute

both scheduling approaches using the same sets of print jobs, printer availability, and staffing constraints. Verify that the framework produces comparable results for the three specified metrics. | *Context:* Discrete-event simulation and comparative scheduling performance. | *Priority:* Must Have

Deliverables

- **Deliverable:** A production-ready, cloud-hosted web application allowing users to upload models, preview settings, monitor queue status, and view real-time print telemetry video feeds.
    - **Requirements:** FR-01, FR-02, FR-03, FR-04, FR-16, NFR-05
- **Deliverable:** A lightweight, containerised Edge Relay daemon capable of being deployed on local maker space networks to interface with physical printers.
    - **Requirements:** FR-12, FR-13, NFR-02
- **Deliverable:** A developer documentation repository including REST API specifications, system architecture diagrams, and a local deployment guide for Arch Linux and macOS environments.
    - **Requirements:** NFR-04

Standards

- **OAuth 2.0 Authorisation Framework (RFC 6749):**
    - *Applicability:* Required for securely authenticating users via Purdue Hackers ID credentials without handling passwords.
    - *Impact:* Dictates the structure of our authentication routing and JWT token verification processes.
    - *Compliance:* We will utilise industry-standard authentication libraries to ensure token exchange strictly adheres to the protocol. URL: https://datatracker.ietf.org/doc/html/rfc6749
- **ISO/ASTM 52900:2021 (Additive Manufacturing — General Principles — Terminology):**
    - *Applicability:* Standardises the nomenclature used across the platform's user interface and documentation.
    - *Impact:* Ensures terms like "build volume," "infill," and "toolpath" are used correctly, preventing confusion for engineering students.
    - *Compliance:* All front-end copy and user-facing documentation will be reviewed against the standard glossary. URL: https://www.iso.org/standard/74514.html
- **WCAG 2.1 (Web Content Accessibility Guidelines):**
    - *Applicability:* Ensures the web portal is fully usable by students with visual or motor impairments.
    - *Impact:* Dictates our UI colour contrast ratios, keyboard navigation logic, and screen-reader compatibility.

- *Compliance:* Front-end development will incorporate automated Lighthouse accessibility audits within the CI/CD pipeline. URL: https://www.w3.org/TR/WCAG21/

GenAI Statement

We did not require AI to write it up, but we did use AI against the requirements to ensure fulfillment in our writeup.